Data Mining

Rule Based Classification

Rule Based Classification

- A technique that uses a set of IF-THEN rules to categorize data. It's highly valued for its simplicity and transparency (explainability).
- Rules are expressed as IF-THEN statements.
  - IF (Condition) THEN (Conclusion)
- Antecedent (IF Part): The condition part, typically a conjunction (AND) of attribute tests.
  - *Example:* (Age = 'Youth') ∧ (Student = 'Yes')
- Consequent (THEN Part): The predicted class label.
  - *Example:* THEN (Buys Computer = 'Yes')
- Complete Rule Example: IF (Age = 'Youth') ∧ (Student = 'Yes') THEN (Buys Computer = 'Yes')

Key Concepts for Rule Assessment

- Coverage: The fraction of training records that satisfy the rule's antecedent. A high coverage means the rule applies to a large portion of the data.
- Accuracy (Confidence): The fraction of records that satisfy the antecedent *and* are correctly classified by the consequent.
  - $\text{Accuracy} = \frac{\text{Number of correct classifications}}{\text{Number of records covered by the rule}}$
- Mutually Exclusive Rules: A set of rules where no two rules are triggered by the same record. Rule-based classifiers often use non-mutually exclusive rules, leading to potential conflicts.
- Exhaustive Rules: A set of rules that covers *all* possible combinations of attributes, meaning every record is covered by at least one rule.

Rule Induction (Learning Rules)

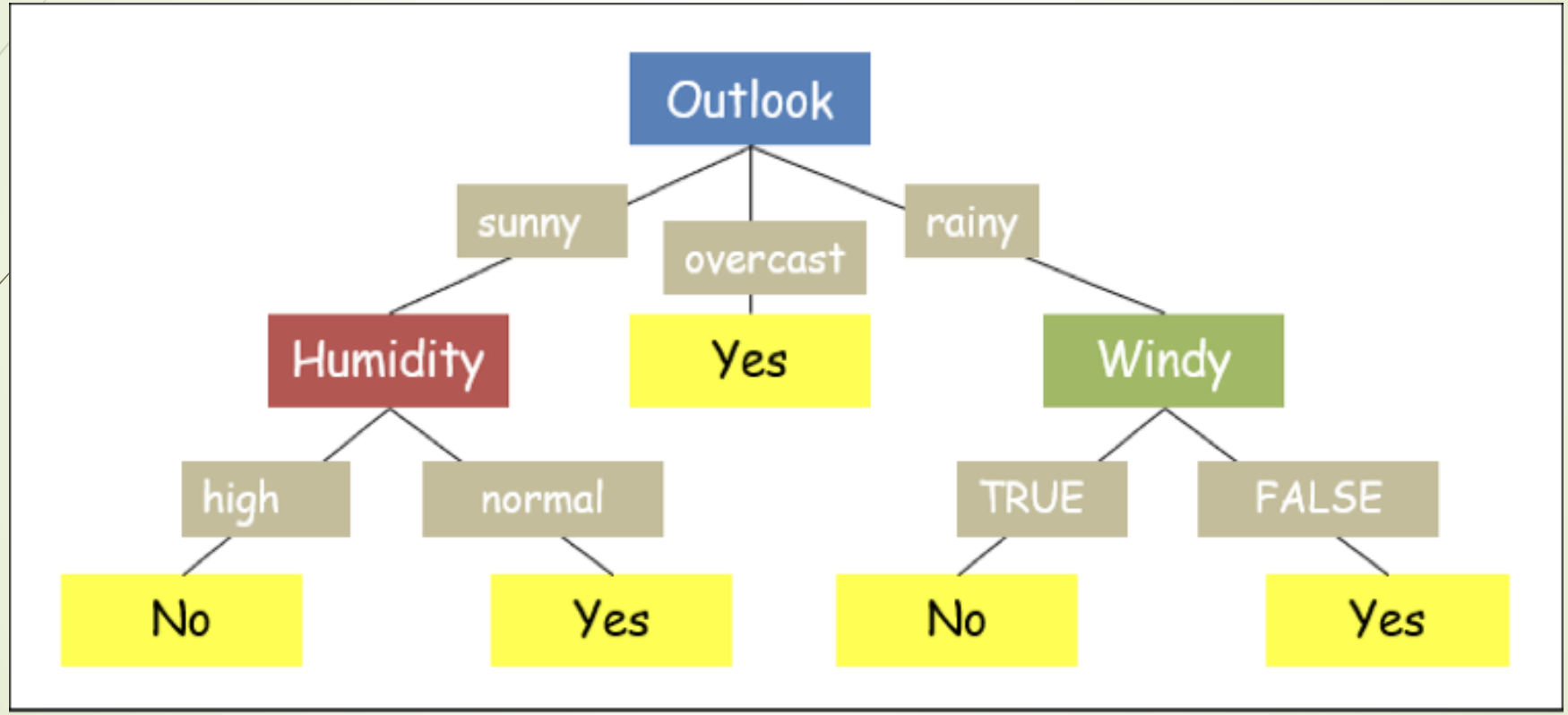
- Direct Method (Sequential Covering Algorithm):
  - Learns one rule at a time for a specific class.
- Indirect Method (From Decision Trees):

Decision Tree Example

Day	outlook	temp	humidity	windy	play
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

https://www.enjoyalgorithms.com/blog/decision-tree-algorithm-in-ml

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